



Client Segmentation

Advanced clustering techniques for personalized banking strategies

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Exploratory Data Analysis

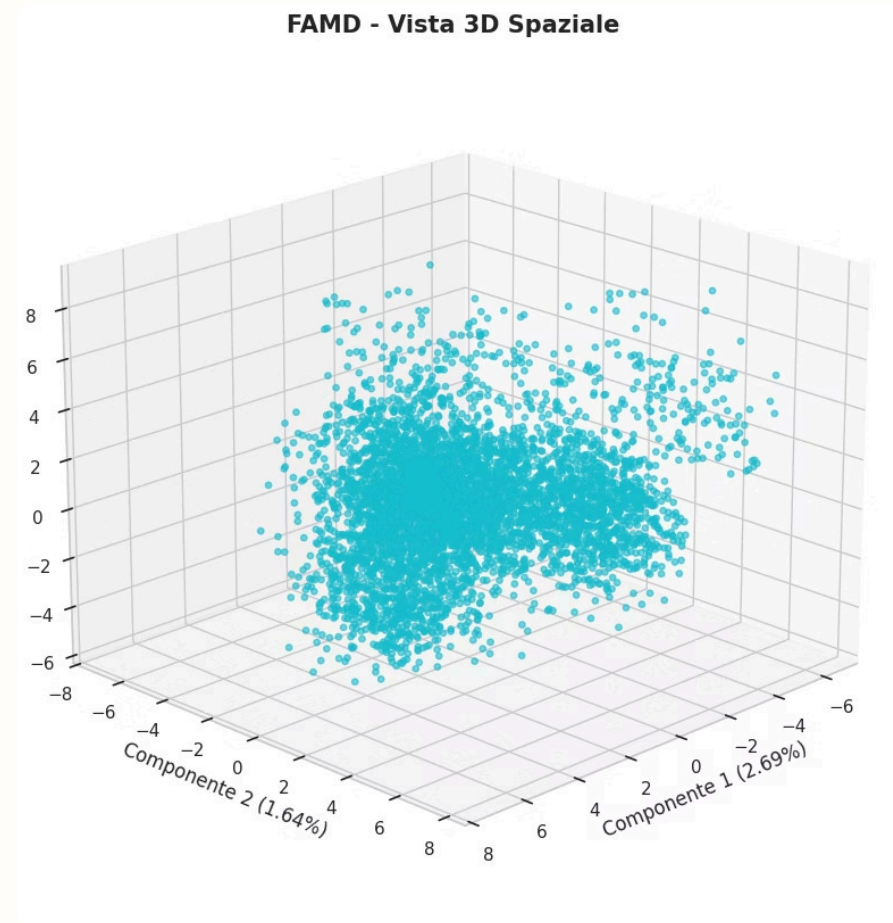
FAMD-3D Analysis

Moderate and weak relationships between all variables: Retaining all features ensures no critical information is lost

Key Features Identified

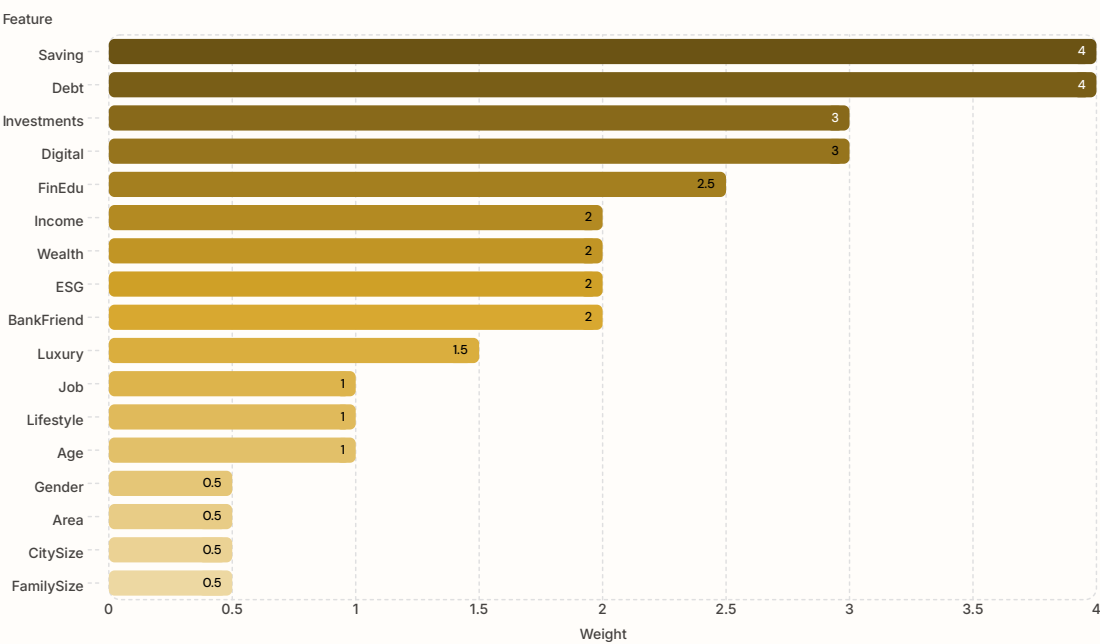
extracted from FAMD and relevant for segmentation:

- Debt
- Digital
- FinEdu
- Investment



Feature Weighting Strategy

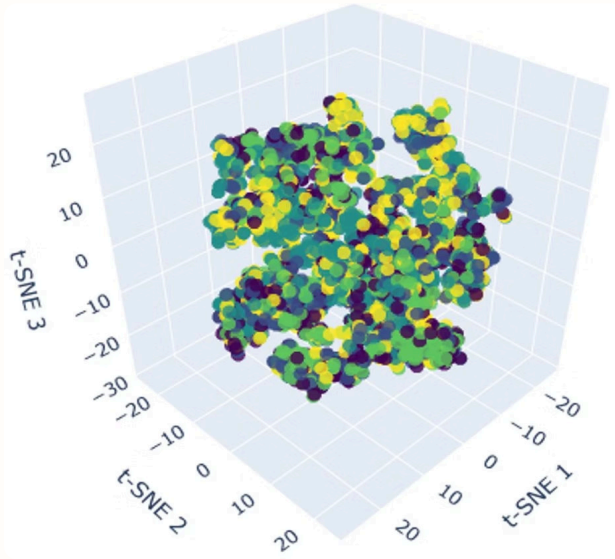
Critical features received higher weights to drive segmentation accuracy.



Data type	Distance Metric	Rationale for Choice
Numerical	Manhattan (L1)	Robustness to outliers.
Categorical	Dice /Jaccard	Focus on shared traits.

Clustering Methodology

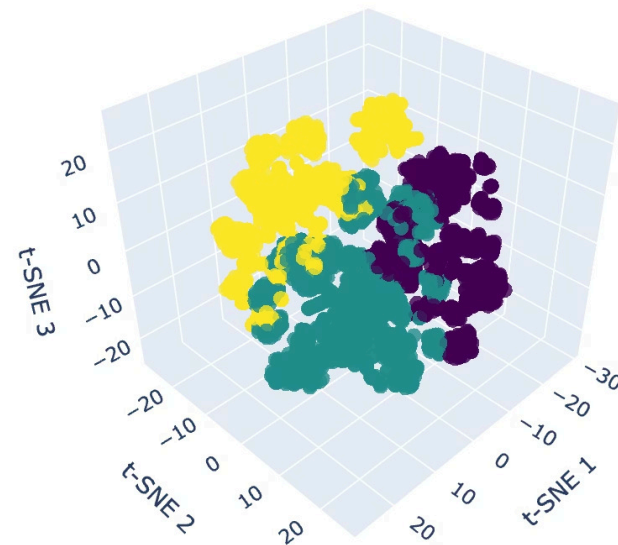
K-Prototypes (k=5)



Conclusion: No clear natural boundaries of clusters. It's needed to use another clustering technique.

DBSCAN Clustering (k=3)

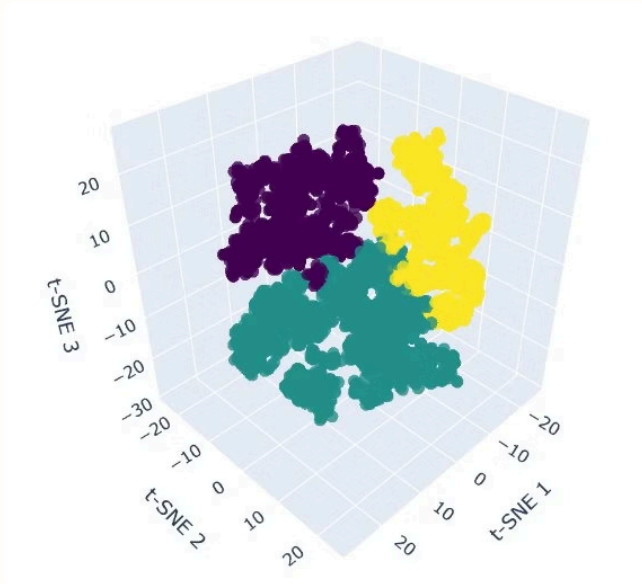
Tested Dice-Manhattan distance metrics



Clustering Methodology

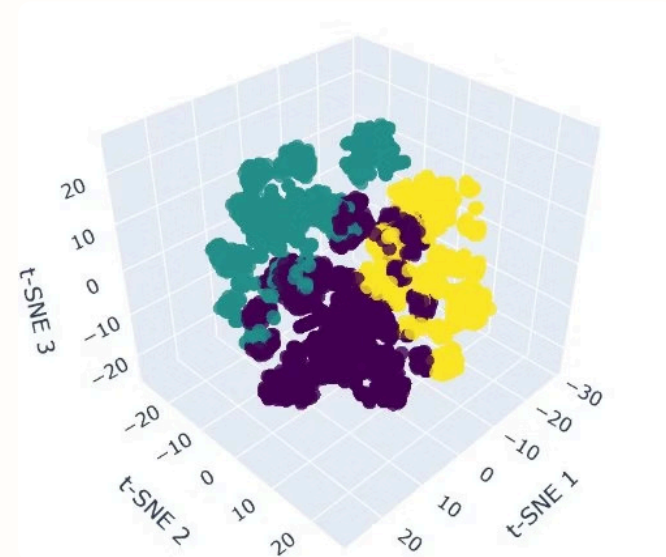
Agglomerative Clustering (k=3)

Tested Dice-Manhattan distance metrics



Agglomerative Clustering (k=3)

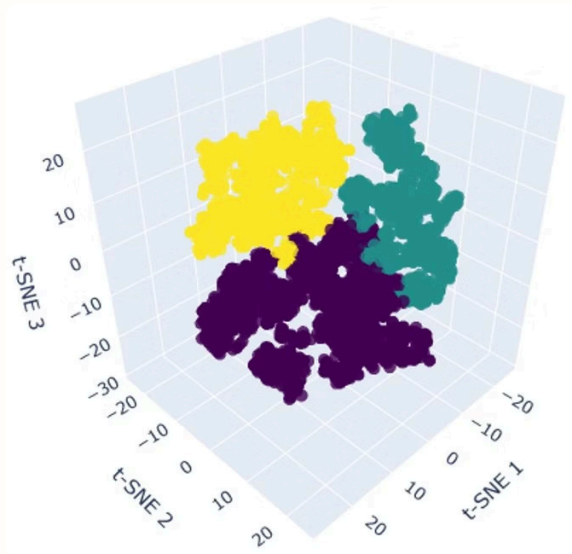
Tested Jaccard-Manhattan distance metrics



Clustering Methodology

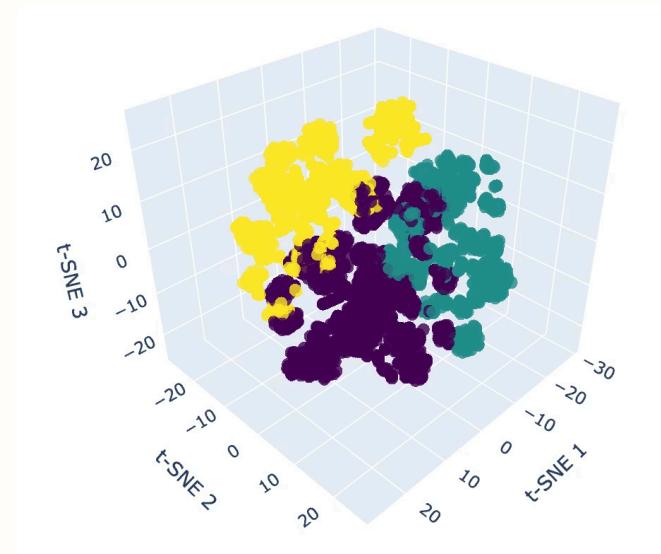
K-Medoids (k=3)

Tested Dice-Manhattan distance metrics



K-Medoids (k=3)

Tested Jaccard-Manhattan distance metrics



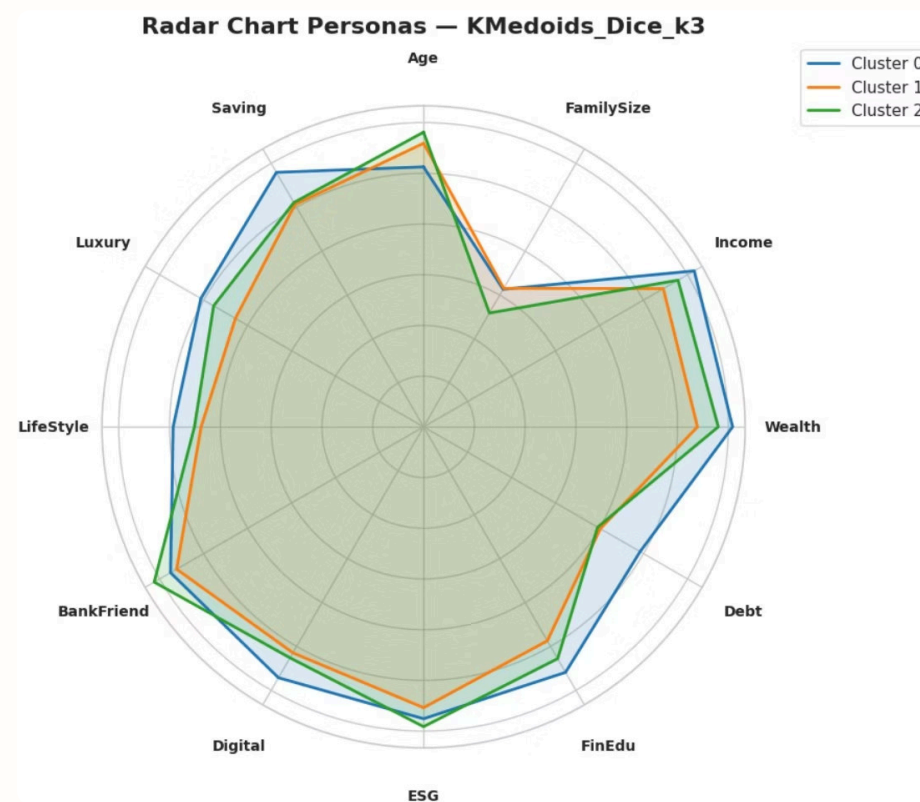
The k=3 Challenge

High Overlap Problem

Radar chart analysis revealed extreme visual similarity across all three clusters. Nearly identical behavioral profiles across 12 dimensions showed:

- High age in all clusters
- High income and wealth uniformly
- Medium debt and FinEdu levels
- High saving capacity everywhere

Result: No distinct, actionable personas possible

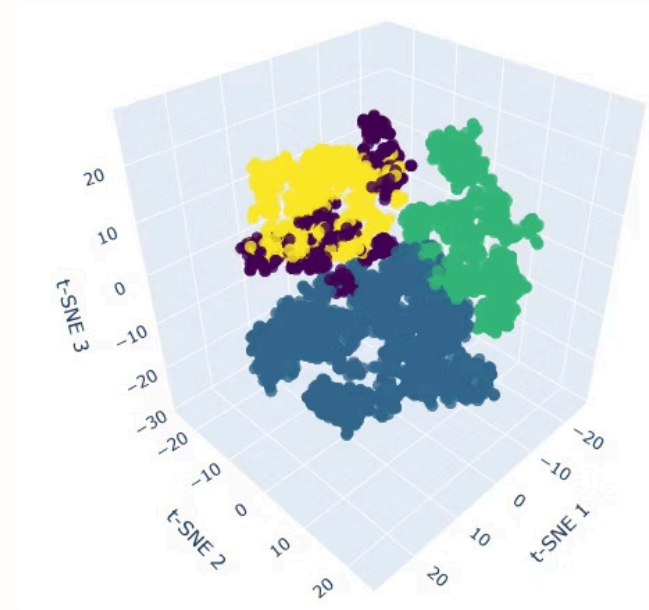


The $k=4$ Solution

Business Manipulation

$k=4$ provides the optimal balance between separation and interpretability. It preserves clear behavioral distinctions across clusters while avoiding the overlap and loss of actionable insight seen at $k=3$.

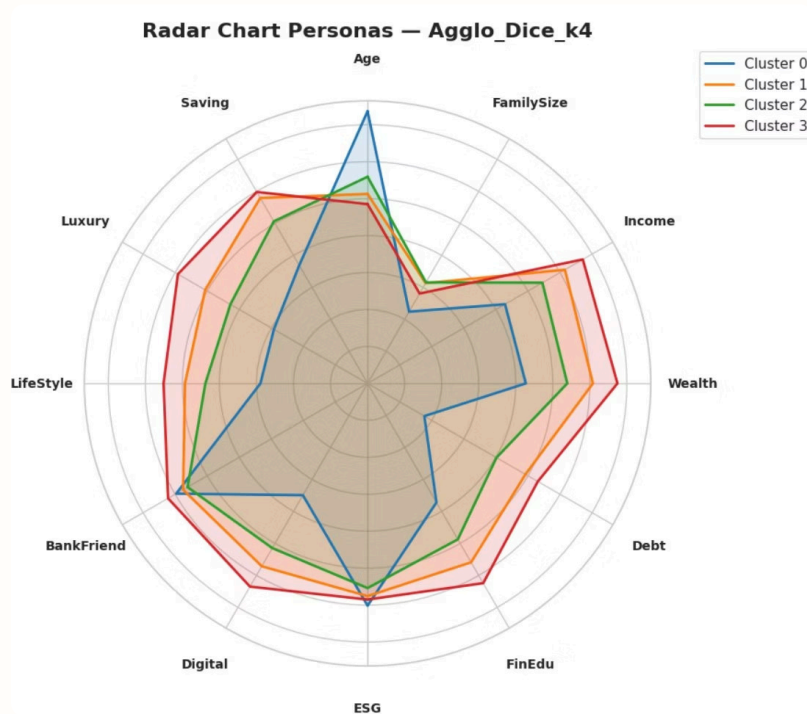
This configuration creates distinct, meaningful personas that are easier to analyze, compare, and activate with tailored strategies.



Clustering visualization showing four clearly separated segments with improved differentiation across the key behavioral dimensions.

Four Distinct Macro-Personas

Moving to k=4 revealed clear, actionable segmentation



Cluster 0: Senior Traditionalists

11% of clients



Cluster 1: Active Families

44% of clients



Cluster 2: Cautious Disinvested

26% of clients



Cluster 3: Premium Affluents

19% of clients

Features	Cluster 0 (11%)	Cluster 1 (44%)	Cluster 2 (26%)	Cluster 3 (19%)
Age & Gender	75–80 – Pred. Female	40–60 – Mixed	60–66 – Mixed	50–60 – Pred. Male
Job & FamilySize	Retired – 2	Employee – 3/4	Employee – 3	Employee/Manager – 1/2
Income	Low-Medium	Medium-High	Medium	Highest
Wealth	Low-Medium	Advanced acc.	Medium	Highest
Debt Level	Very Low	Moderate-High	Medium-Low	Medium-High
Saving Capacity	Low	High	Medium	Highest
Investments	Lump sum	Gradual acc.	None	Large lump sum
Financial Education	Low	Medium-High	Low	Highest
Digital Propensity	Low	Medium-High	Medium	Highest
Bank Relationship	High	Good	Lowest	High
ESG	High	Medium	Low	Medium

Persona Deep Dive: Traditionalist & Family



Laura – Senior Traditionalists

- **Profile:** A **75-year-old** retiree whose primary goal is capital preservation. He has a strong aversion to risk and is completely unfamiliar with digital tools. The bank is a physical place made of human relationships;
- **Key Behaviors:**
Digital & Risk: Minimum
Bank Trust: Maximum
Investments: Rare, safe lump-sums
- **Business Strategy:** Focus strictly on face-to-face relational advisory. Avoid pushing digital platforms. Propose highly secure, ESG-compliant, and low-risk capital preservation solutions.



Frank – Active Families

- **Profile:** A **58-year-old** active worker managing family commitments. She is independent, tech-savvy, and financially educated. Her goal is long-term stability—growing savings gradually and safely rather than taking massive risks.
- **Key Behaviors:**
Digital Propensity: High & Autonomous
Saving Capacity: Continuous & Disciplined
Investments: Periodic Accumulation (PAC)
- **Business Strategy:** The ideal target for long-term financial planning. Offer simple, automated digital accumulation tools (PAC) seamlessly integrated with on-demand advisory support for family budget management.

Persona Deep Dive: Cautious & Premium



Giulia – Cautious Disinvested

- **Profile:** A 61-year-old approaching retirement with stable income, but defined by **financial inactivity**. She leaves liquidity stagnant in her account to avoid risk. This stems from low financial literacy and, crucially, **low trust in the bank**.
- **Key Behaviors:**
 - Investments:** Zero (Liquidity parked)
 - Bank Trust:** The lowest panel
 - Core Emotion:** Fear of complexity
- **Business Strategy:** Do not push products immediately. The strategy must focus on **rebuilding trust and transparency**. Guide her through basic financial education toward zero-surprise, entry-level investments



Marco – Premium Affluents

- **Profile:** A 55-year-old established professional with high, actively managed wealth. He is highly digitalized, makes autonomous decisions, and expects fast, premium services. He combines a luxury lifestyle with a solid capacity to maximize returns.
- **Key Behaviors:**
 - Wealth & Income:** Maximum
 - Digital & FinEdu:** Advanced / Autonomous
 - Investments:** Large-value lump sums
- **Business Strategy:** Standard retail offers will fail here. Marco requires **highly customized wealth management**, a strictly premium service model, and cutting-edge digital platforms paired with expert advisory

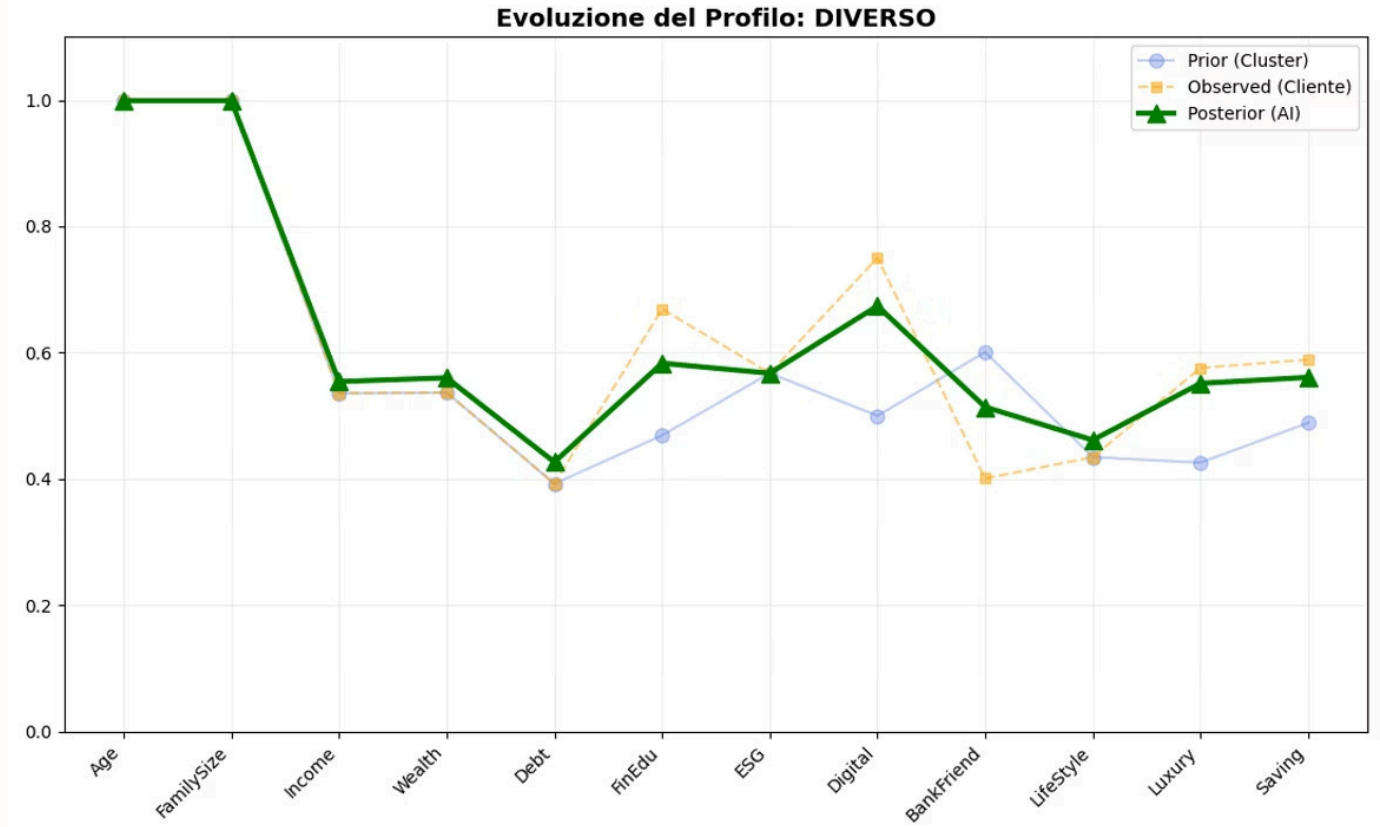
Dynamic Persona Evolution

Bayesian Updating Framework

Personas evolve beyond static segments through mathematical adaptation:

1. **Prior:** Original cluster profile
2. **Observed:** Individual customer behavior
3. **Posterior:** AI-generated balanced update

Multivariate structural updates recalibrate deep relationships between all variables dynamically



Thank you
for your
time and
attention

